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Solidification of Undercooled Melts of Al-Based Alloys on Earth and in Space

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Containerless processing of droplets and drops by atomization and electromagnetic levitation are applied to undercool metallic melts and alloys prior to solidification. Heterogeneous nucleation on crucible walls is completely avoided giving access to large undercoolings. Experiments are performed both under terrestrial (1 g) conditions and in reduced gravity (μg) as well. Microgravity conditions are realized by the free fall of small droplets during atomization of a spray of droplets, individual drops in a drop tube and by electromagnetic levitation of drops during parabolic flights, sounding rocket missions, and using the electro-magnetic levitator multi-user facility on board the International Space Station. The comparison of both sets of experiments in 1 g and μg leads to an estimation of the influence of forced convection on dendrite growth kinetics and microstructure evolution.

INTRODUCTION

Metallic materials are prepared from the liquid state as their parent phase. To date, efforts are directed toward virtual material design with computer-assisted modeling. Computational materials science from the liquid state requires detailed knowledge of crystal nucleation and crystal growth. Both processes are driven by an undercooling of the liquid below its equilibrium melting temperature. At large undercoolings, this gives access to nonequilibrium solidification pathways, which can form metastable solids, that may differ in their physical and chemical properties from their stable counterparts.

To achieve the state of a deeply undercooled melt, heterogeneous nucleation has to be reduced. Turnbull used the method of volume separation of heterogeneous sites by sample subdivision into many small particles to isolate the heterogeneous motes in a few particles.[1] Perepezko refined this technique by subdividing the macroscopic melt into small particles suspended in an inert carrier fluid. The carrier fluid acts as an agent removing heterogeneous motes on the surface of the small droplets as well.[2] Even undercooling of macroscopic melts in a fluxing agent can be used to achieve huge undercoolings, enabling the formation of bulk glass formation of metallic alloy.[3,4] During the last few decades, levitation techniques have been progressively used to produce liquid drops of metals, alloys, and semiconductors. These techniques offer the additional benefit that a freely suspended drop is directly accessible for in situ diagnostics to determine thermophysical properties and investigate solidification far away from thermodynamic equilibrium.

In the present article, an overview is given by the partners of the NEQUISOL* project funded by the European Space Agency. Solidification is studied under the conditions of containerless processing on

*Abbreviation of "Non-EQUIlibrium SOLidification, Modelling for Microstructure Engineering of Industrial Alloys".

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Earth and in reduced gravity.[5] A sophisticated impulse atomization facility is used to produce a spray of droplets. Atomization is suitable for investigating the statistics of the formation of various structures and microstructures as a function of droplet size and cooling rate. Nevertheless, these techniques do not allow for direct diagnostics of individual droplets. For this purpose, we apply electromagnetic levitation on Earth and in reduced gravity.[6] Comparing results obtained under terrestrial and reduced gravity conditions allows for the investigation of fluid flow effects in dendrite growth dynamics and for the microstructure evolution of solidification. The experiments are accompanied by the development and application of mesoscopic models to analyze solidification including fluid flow by forced convection.[7]

EXPERIMENTS

In the present project, methods of containerless processing are applied to undercool metallic melts without any contact with a solid crucible. That allows for deep undercoolings and solidification far away from equilibrium. These techniques include gas atomization, impulse atomization drop tube, melt fluxing, and electromagnetic levitation both on Earth and in Space. Most of the experimental techniques used within the NEQUISOL project are described in detail elsewhere.[8]

Levitation experiments allow for measurements of temperature-time profiles during an entire undercooling and solidification experiment. The strong electromagnetic field needed to compensate the gravitational force causes strong stirring effects in the liquid. These are essentially reduced if the electromagnetic levitation technique is applied in reduced gravity. In the environment of space, the forces to compensate disturbing accelerations are of some orders of magnitude smaller compared with experiments on the ground. A special instrument called TEMPUS** has been designed to provide means of containerless processing in space[9]. Opposite to levitation on Earth, positioning and heating are separated in TEMPUS by placing the sample into the superposition of a quadrupole and a dipole field.[10] The quadrupole field serves as positioning and the dipole field for heating. This two-coil concept leads to a drastic improvement of the overall efficiency and allows for independent control of positioning and heating. The concept is illustrated in Fig. 1. It shows the two-coil systems together with a pyrometer to measure contactless the temperature and a high-speed camera to record the rapid propagation of dendrites. The dendrite growth velocity is measured as a function of undercooling.

**TEMPUS is a German acronym for *Tiegelfreies Elektro-Magnetisches Prozessieren Unter Schwerelosigkeit*.

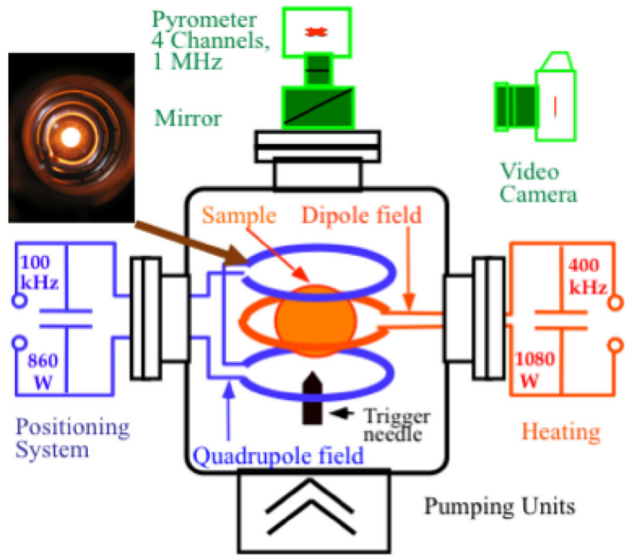

Fig. 1. Schematic view of the TEMPUS facility. The heart of this facility consists of a quadrupole field for positioning and a diploe field for efficient heating. Experiments can be conducted under ultrahigh-vacuum (HV) conditions in contrast to EML on Earth.

MODELING

A segregation model was developed to follow the transformation path of the alloy solidified by a containerless process, such as levitation or atomization. Because such an initial condition is more generally valid for the formation of equiaxed grain structures, it is of major technological relevance for all casting technologies. The model is presented in detail elsewhere.[11,12] It is based on an extension of previously introduced concepts.[13,14] A global heat balance (assuming uniform temperature) coupled with total mass and solute mass balances (assuming constant and equal density of all phases) are considered. The system is made of an initial spherical liquid droplet at the alloy composition. Figure 2 gives a schematic view of the solidification sequence. It successively includes cooling of the lone liquid $l^{(0)}$ that composes zone (0), growth of a mushy zone (1) after nucleation of the primary dendritic solid phase $s_1^{(1)}$, and growth of a eutectic zone (2) after nucleation of a structure $s_2^{(2)}$ The later eutectic structure develops first in place of the interdendritic liquid $l^{(1)}$ and then in place of the remaining liquid $l^{(0)}$. Finally, cooling of the fully solid system takes place.

The growth kinetics of the envelope of the mushy zone (1) and eutectic zone (2) are computed using sharp interface models of dendrite growth kinetics[15,16] and the Jackson-Hunt eutectic theory,[17] respectively. The interdendritic liquid $l^{(1)}$ located in the mushy zone is characterized by a high solute composition. It exchanges solute with the poorer extra-dendritic liquid $l^{(0)}$ located outside the mushy zone. Diffusion in the primary solid phase s_1 is also accounted for. The model is fully coupled with a thermodynamic equilibrium calculation.[18]

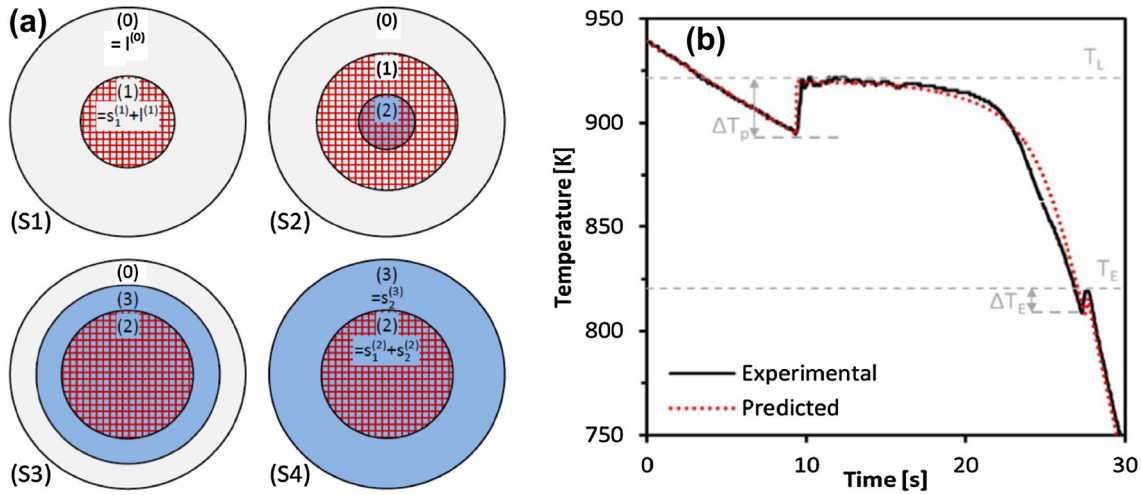

Fig. 2. (a) Schematized solidification sequence of a binary eutectic alloy (see text); (b) temperature-time profile measured by electromagnetic levitation and reproduced by modeling segregation.

A sharp interface model[19,20] has been recently extended to describe the growth dynamics of anisotropic dendrites with forced convection as a function of undercooling.[21,22] The present model takes into account a change of the liquidus line and the slope of the liquidus line at high undercoolings because of kinetic reasons. In addition, influences of convection are integrated by introducing Peclet numbers for fluid flow and its impact on thermal and mass transport. These are mandatory to analyze the dendrite growth dynamics under the condition of forced convection as present in electromagnetic levitation experiments on Earth.

Phase-field modeling of microstructure evolution is based around an adaptive mesh multigrid solver for the coupled thermo-solutal system.[23,24] The model is distinct from most other phase-field models as we do not require growth to be in an isothermal system, nor in an imposed temperature gradient, but rather the model solves the coupled equation set for the rejection of solute and latent heat into the liquid ahead of the growth front. As the ratio of thermal diffusivity, α, to solute diffusivity, D (termed the Lewis number, Le), is typically $\gg 1$, a range of advanced numerical methods are employed to cope with the resulting multiscale problem. These include:

1. Adaptive meshing refinement: This allows mesh cells to be located only in regions where the phase field, or one of the other fields, is varying significantly, typically close to the solid-liquid interface. The mesh refinement level is controlled by the gradients of the phase, solute, and thermal fields.
2. Implicit time-stepping: Mesh adaptivity allows a highly refined mesh to be employed while retaining a moderate total element count. Nevertheless, for the explicit forward Euler scheme employed in most phase-field codes, this leads to restrictively small time-steps resulting from a stability criterion of the form $\Delta t \propto (\Delta x)^2$. This is overcome by using an implicit (second-order backward difference) time-stepping scheme that is unconditionally stable. Yet, at each time-step, a large, but space, matrix system needs to be inverted.
3. Multigrid solver: Multigrid solvers are the most efficient means of solving a large sparse system of equations, with the computational load scaling as $O(N)$, where N are the number of degrees of freedom. This compares with $O(N^2)$ for Gauss elimination or $O(N^{3/2})$ for successive over-relaxation or conjugate gradient methods.
4. Parallel execution: The system is built on the PARAMESH platform for execution on modern parallel HPC architectures.

The combined application of these numerical methods means that we operate the phase-field model at Lewis numbers approaching realistic values[25] and with a diffuse interface width.

EXPERIMENTAL RESULTS AND DISCUSSION

Figure 3 shows the dendrite growth velocity as a function of undercooling of the $Al_{50}Ni_{50}$ intermetallic compound. Both results from terrestrial and experiments in reduced gravity are shown.[26] The green symbols represent measurements by using electromagnetic levitation on Earth with strong forced convection. The triangles, squares, and pentagons give results of equivalent measurements in reduced gravity measured using the TEMPUS facility during several parabolic flights. The dashed line shows the results of calculation within sharp interface theory neglecting convection, whereas the solid line gives the corresponding results if fluid flow motion with velocities up to U_o = 0.6 m/s are

assumed comparable with magnetohydrodynamic computations[27] and experiments[28] for electromagnetically levitated drops on Earth.

The growth behavior of $Al_{50}Ni_{50}$, $Al_{55}Ni_{45}$, $Al_{60}Ni_{40}$, $Al_{65}Ni_{35}$, $Al_{68.5}Ni_{31.5}$, $Al_{70}Ni_{30}$, and $Al_{75}Ni_{25}$ alloy was experimentally investigated. The results are shown in Fig. 4. In the case of $Al_{50}Ni_{50}$ and $Al_{55}Ni_{45}$, the dendrite growth velocity increases monotonically with undercooling. An increase of Al-concentration by 5 at.% leads to an essential change of the dendrite growth characteristics. A decrease of growth velocity is observed at undercoolings $\Delta T < 175$ K. The growth velocity passes a minimum and subsequently increases with rising undercooling. If the concentration of Al is further increased, the minimum is shifted toward higher undercoolings.[29]

To investigate a possible influence of forced convection of the anomalous growth behavior of Al-rich Al-Ni alloys, the Raney-type alloy $Al_{68.5}Ni_{31.5}$ was the subject of various experiments in reduced gravity during TEXUS sounding rocket missions TEXUS 44[30] and TEXUS 49.[31] Figure 5 comprises the results of these investigations in direct comparison with results from 1 g investigations.

Although the initial results of TEXUS 44 mission (red triangles) are different compared with the results obtained on Earth (1 g), the results of TEXUS 49 mission (cyan diamonds) do not indicate a significant difference of both comparative sets of experiments. Nevertheless, they are located in the regime where the growth velocity is larger than the estimated fluid flow speed $U_s \approx 0.3$ m/s induced by the electromagnetic fields in 1 g (cf. dashed line).[27] The experiments of the TEXUS 44 mission suffer from a low measuring frequency of the camera. The measuring frequency was enhanced by a factor of 4 during the TEXUS 49 mission. A reevaluation of the measurements on TEXUS 44 leads to values indicated by the green diamonds.

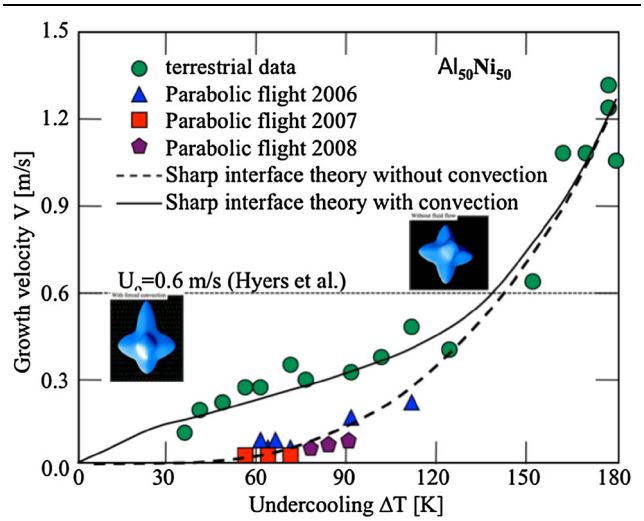

Fig. 3. Dendrite growth velocity V as a function of undercooling measured of $Al_{50}Ni_{50}$, both under terrestrial (green circles) and μg conditions during parabolic flight missions using the TEMPUS facility (triangles, squares, pentagons). The dashed line gives the prediction of dendrite growth theory neglecting fluid flow, whereas the solid line represents the prediction of dendrite growth theory assuming a fluid flow velocity of U_o = 0.6 m/s as determined by magneto-hydrodynamic simulation.

The insert of Fig. 5 shows a significant difference of the microstructure of an $Al_{68.5}Ni_{31.5}$ sample solidified at an undercooling ΔT = 80 K under 1 g (left) and in μg (right). The microstructure on the left exhibits grains without porosity, whereas the microstructure on the right reveals great porosity in the volume and dendrites, which grow from a surface layer isotropically to the center. From these observations, it is concluded that solidification in reduced gravity may consist of two steps. First, a layer on the surface is solidifying that acts as heterogeneous nucleation site for dendrites that grow to the center of the sample.

NEQUISOL contributes two samples of $Al_{40}Ni_{60}$ to batch 1 of the EML experiments on board the ISS that were recently processed performing 25 solidification cycles successfully, 23 cycles for sample 1 and 2 cycles for sample 2. Both samples are of equal nominal compositions. Using sample 2, external triggering was successfully applied to initiate solidification at small undercoolings. Figure 6 shows the distribution of the nucleation undercooling measured. The average undercooling is inferred to be around ΔT = 100 K. That is by about 70 K smaller than the maximum undercoolings measured for the same alloy by electromagnetic levitation under terrestrial (1 g) conditions. So far it is not yet understood where this difference in maximum undercooling is coming from. There may be different effects that could be considered to explain this finding. The forced convection present in 1 g experiments will hinder heterogeneous nucleation of a specific origin. In alloys, the growth of an embryo to a nucleus of critical size is diffusion controlled. The atomic diffusion speed is in the order of 10 m/s. Hence, it is unlikely that this effect explains the difference in nucleation undercooling in 1 g and μg.

The maximum undercoolings as measured in both 1 g and μg are significantly smaller than estimated for the physical limit of the undercoolability as given by the onset of homogeneous nucleation. According to work by David Turnbull,[32] the nucleation limit in the presence of homogeneous nucleation should be in the order of $\Delta T_{max} \approx 0.2$ T_L = 374 K with T_L = 1870 K the liquidus temperature of $Al_{40}Ni_{60}$ alloy. Therefore, there must be a specific heterogeneous nucleation side being active in the levitation experiments.

In the case of Al-based alloys, it is well known that there is a strong tendency of Al in the liquid state of the alloy to form Al_2O_3 oxide. This oxide is stable with a melting temperature of $T_L \approx 2350$ K and a boiling temperature of about 3225 K. If processing the Al-Ni alloy in an environmental gas

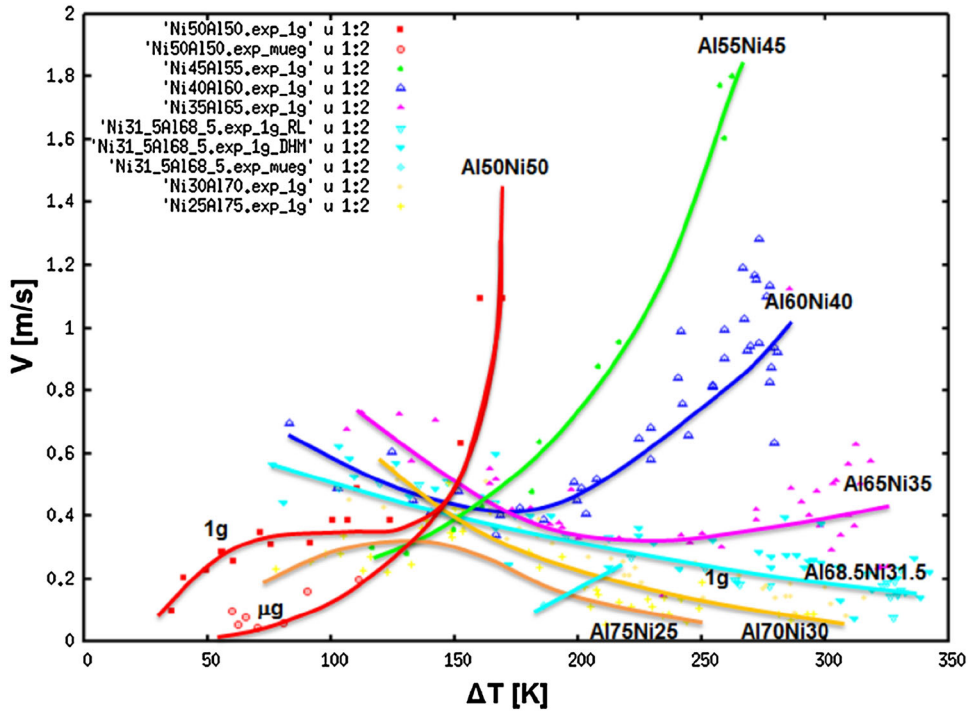

Fig. 4. Dendrite growth velocity V as a function of undercooling ΔT of variously concentrated Al-Ni alloys. The lines show data trends as a guide to the eye.

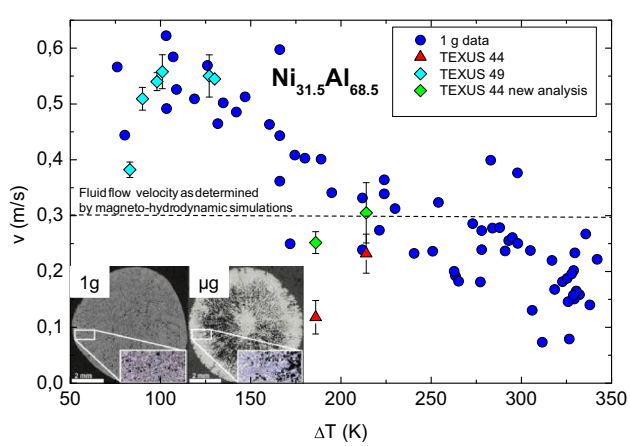

Fig. 5. Dendrite growth velocity V versus undercooling ΔT of $Al_{68.5}Ni_{31.5}$ alloy, measured on Earth (blue dots) and in reduced gravity during TEXUS 44 mission (red triangles) and TEXUS 49 mission (light blue diamonds). The insert shows the microstructures of $Al_{68.5}Ni_{31.5}$ solidified at ΔT = 85 K on Earth (left) and in reduced gravity at comparative undercooling (right) (Color figure online).

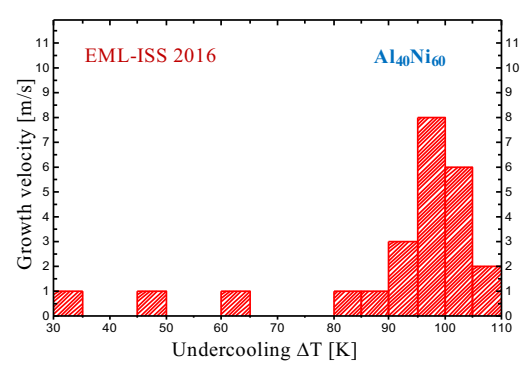

Fig. 6. Distribution of nucleation undercooling as measured on $Al_{40}Ni_{60}$ alloy by electromagnetic levitation (EML) in reduced gravity on board the International Space Station (ISS).

atmosphere, a solid Al_2O_3 layer is formed at the surface of the sample during heating within a time interval that is short compared with the processing time. In the present experiments, the $Al_{40}Ni_{60}$ sample is processed in an Ar atmosphere of nominal purity of 6 N at a pressure of 350 mbar. If one assumes a reaction probability of 1 (each individual reaction $4Al + 3O_2 = 2Al_2O_3$ is successful), a single surface layer of Al_2O_3 is formed at a pressure of 10^{-6} mbar within a time interval of 1 s. Under the present experimental conditions, the partial pressure of oxygen is in the order of 10^{-4} mbar. Therefore, the time for the formation of an Al_2O_3 surface layer is in the order of 10^{-2} s and, hence, is much smaller than the processing time in the order of several seconds. At temperatures $T > 730$ K. a surface layer of Al_2O_3 is progressively formed that can act as an effective heterogeneous nucleation side. If the Al_2O_3 would be present during cooling, the liquid sample will crystallize immediately if the temperature is falling below the liquidus temperature because of the high catalytic potency of Al_2O_3 for the formation of a heterogeneous nucleus of critical size.

There are chemical reactions that transfer the Al_2O_3 to AlO and Al_2O as investigated by Hoch and Johnston.[33] These are the following transformations:

$$4Al + Al_2O_3 \rightarrow 3Al_2O \quad \text{at temperatures } 1273\,K < T < 1573\,K$$
$$1Al + Al_2O_3 \rightarrow 3AlO \quad \text{at temperatures } T > 1873\,K$$

Al_2O and AlO are highly volatile as shown by the same authors. An AlO sample in mass of 15 mg completely evaporates within 1 h at a temperature of 2150 K. Therefore, a surface layer of Al_2O_3 in thickness of, e.g., 10 nm may disappear within a time interval of a few seconds if the melt is heated up to 2170 K prior to cooling. Such a time interval is comparable with the experimental time for a heating and solidification cycle in the EML.

There is another transformation of Al_2O_3 that merits attention in the present discussion. Crystalline Al_2O_3, c-Al_2O_3, transforms to amorphous Al_2O_3, a-Al_2O_3 in the temperature regime 1000 K < T < 1300 K. As both solid phases have different mass densities of 3.2 g/cm^3 in the case of c-Al_2O_3 and 3.98 g/cm^3 in the case of a-Al_2O_3, a solid Al_2O_3 layer may crack upon heating an Al-Ni sample forming nanosized Al_2O_3 clusters that can dissolve into the liquid sample and can act as heterogeneous nucleation sides within the volume of the liquid alloy.[34] The curvature of the nm particles determines the catalytic potency of these heterogeneous nucleation modes.[35] The dissolution of the particles with respect to their distribution inside the liquid depends on the fluid flow motion and can affect the maximum undercooling and may be one reason for the different undercoolings measured in 1 g and μg. Further on, also the liquid of the metallic melt in contact with the solid layer of Al_2O_3 develops an ordering within some interatomic distances as evidenced by investigations on undercooled pure Al.[36] This ordering affects the nucleation undercooling as discussed by Greer.[37] It can be speculated that this ordering mechanism depends on the fluid flow motion inside the melt and therefore affects the nucleation undercooling. All of these aspects shall be investigated in more detail to understand the different undercoolings of liquid $Al_{40}Ni_{60}$ alloy as measured in 1 g and μg, respectively.

In addition to the nucleation issue, also the dendrite growth velocity kinetics was investigated in the EML-ISS experiments of $Al_{40}Ni_{60}$ alloy. For doing so the concept as described in detail elsewhere has been applied to determine the velocity of the solidification front that propagates through the volume of the sphere-like sample.[38] In the previous experiments, both on Earth and during parabolic flight missions, the growth velocity was directly inferred from the propagation speed of the intersection line of the solidification front with the sample surface. This data are therefore afflicted with a systematic error that is under investigation by reevaluating the previous experiments using the method described in Ref. 38. Figure 7 shows the preliminary evaluation of the dendrite growth velocity as a function of undercooling (red stars) as measured on $Al_{40}Ni_{60}$ alloy during the EML experiments on board the ISS. These measurements are compared with previous results of measurements of the dendrite growth velocity both during parabolic flight missions (blue triangles and red squares) and in the Earth laboratory (green circles). The upper part of the Fig. 7 displays three photographs taken by the high-speed camera at subsequent steps during solidification. The solid part appears brighter than the liquid because the solidified part is heated up as a result of the release of the heat of fusion. According to Fig. 7, no essential difference can be detected for all measurements within the scatter of the data. There is one exception. During a previous parabolic flight mission, the same alloy was processed using the TEMPUS facility. It succeeded to record one value of the dendrite growth velocity at an undercooling of 150 K. This value is significantly smaller than the value of the dendrite growth velocity measured at the same undercooling under terrestrial conditions.

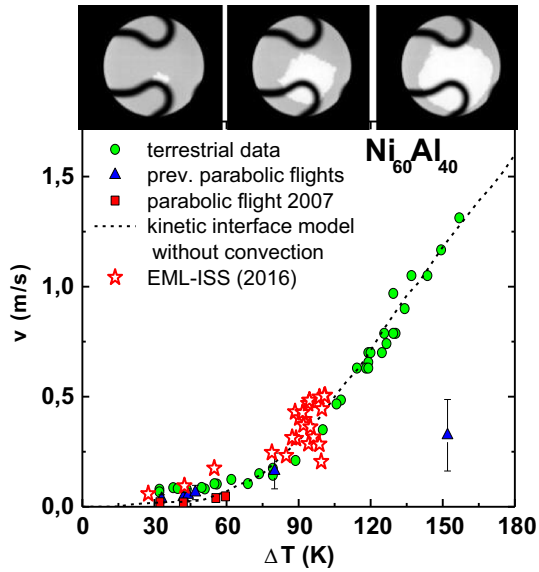

Fig. 7. Dendrite growth velocity V as a function of undercoolig ΔT of $Al_{40}Ni_{60}$ alloy, measured under terrestrial conditions (green circles), during parabolic flight missions (blue triangles, red squares) and on board the International Space Station using the EML (red open stars). The ingots in the upper part display three pictures taken by the high-speed camera at subsequent steps of solidification (Color figure online).

PHASE IDENTIFICATION IN Al-Ni Alloys

AlNi alloys are important to the catalysis industry. The activity of a Raney nickel catalyst is strongly affected by the morphology and weight fraction of Al_3Ni in the structure. $Al_{68}Ni_{32}$ and $Al_{79}Ni_{21}$ were atomized using IA in both helium and nitrogen gases. The microstructure of a range of powder sizes was examined quantitatively using neutron diffraction and SEM.[39–41] Similar

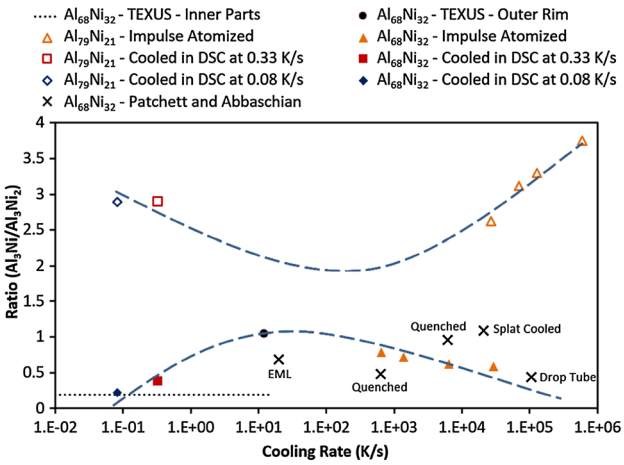

Fig. 8. The Al_3Ni/Al_3Ni_2 ratio as a function of cooling rate for $Al_{68}Ni_{32}$ and $Al_{79}Ni_{21}$ samples produced using IA, DSC (0.08 and 0.33 K/s) and EML during TEXUS 44 campaign ($Al_{68}Ni_{32}$ only), as well as results reported by Patchett and Abbaschian for $Al_{68}Ni_{32}$.[43] The measured cooling rate of the TEXUS sample (Outer Rim) varies from 14 K/s before solidification to 10 K/s after. The inner parts cooling rate (dotted line) is unknown but is less than at the outer rim. The two dashed lines highlight the trends in Al_3Ni/Al_3Ni_2 ratio for the two compositions studied.

compositions were slowly cooled (0.08 and 0.33 K/s) in a differential scanning calorimeter (DSC). Also, an $Al_{68}Ni_{32}$ droplet was processed using the TEXUS sounding rocket that provided 320 s of reduced microgravity. The sample from the TEXUS flight was examined using neutron diffraction and SEM. Identification and quantification of the Al_3Ni_2 and Al_3Ni phases were carried out.

Comparing the phase quantification of Al_3Ni_2 and Al_3Ni for the IA, DSC and TEXUS samples revealed that cooling rate played a major role in the formation of these phases and their phase fractions.

Figure 8 shows the ratio of Al_3Ni/Al_3Ni_2 versus cooling rate for both $Al_{68}Ni_{32}$ and $Al_{79}Ni_{21}$ alloys processed using EML, IA, and DSC.[42] The work of Patchett and Abbaschian is also shown in Fig. 8. For the $Al_{68}Ni_{32}$ alloy, the ratio of Al_3Ni/Al_3Ni_2 appears to decrease with increased cooling rate and then increases again. Also, the ratio is higher than for $Al_{79}Ni_{21}$ alloy and the trend with increasing cooling rate is the opposite. As the cooling rate increases, the ratio of Al_3Ni/Al_3Ni_2 increases and appears to reach a maximum at around 100 K/s. Thereafter the ratio decreases with increasing cooling rate. The results seem to suggest that coarser sized powders may offer the optimum ratio of Al_3Ni/Al_3Ni_2 for catalyst performance.

CONCLUSION

It was demonstrated that solidification kinetics and microstructure evolution are strongly influenced by changes in mass and heat transport resulting from forced convection during experiments under terrestrial conditions. These disturbing effects can be eliminated by careful analogous experiments in a reduced gravity environment as demonstrated for measurements of dendrite growth velocity of undercooled Al-Ni alloy. Highly accurate measurements in reduced gravity are therefore a mandatory requirement to quantitatively verify solidification models in particular in the regime of crystallization velocities being of comparable order of magnitude of the fluid flow velocity induced by forced convection. The upcoming period of the use of the International Space Station opens up a new area of research in which a continuous and long-duration series of experimental investigations in reduced gravity becomes possible.

ACKNOWLEDGEMENTS

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